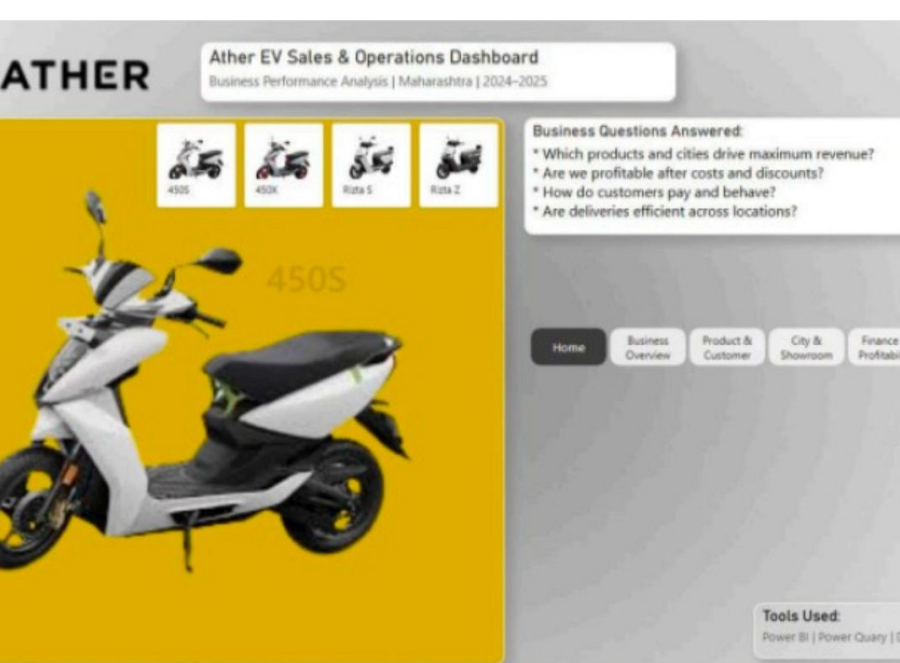




Electric Vehicle Sales & Operations Dashboard

End-to-End Power BI & Data Engineering Solution for Automotive Networks

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The Business Challenge

EV companies face major bottlenecks while scaling – from managing store sales and inventory to tracking loan approvals and delivery delays.

The Dataset

Modelled on real-world EV store transactions, inventory, and telemetry data.

Core Problem

Sales, finance, and logistics data were saved in different systems with no central view.

Our Goal

Build a simple Power BI dashboard to bring all business numbers into one screen.

4 Key Questions This Project Answers

1

Which EV Models Sell Best?

Identifying top revenue-generating models vs. low-performing variants.

2

Which Stores Perform Best?

Comparing sales across different cities and store locations.

3

How Important Is Financing?

Understanding how many customers buy vehicles via EMIs/Loans and its effect on profit.

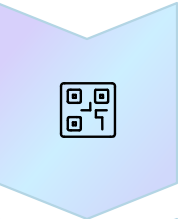
4

Where Are Deliveries Delayed?

Finding out how long it takes to deliver vehicles and where customers face delays.

How The Data Moves — Pipeline Architecture

A clean, five-step pipeline transforms raw transactional data into interactive Power BI reports — from Python-generated records to a fully connected star schema.



Python

Generated clean transactional records for sales, stores, and deliveries



SQL Server

Stored all raw data in structured SQL database tables



Power Query

Cleaned, filtered, and fixed missing values in the data



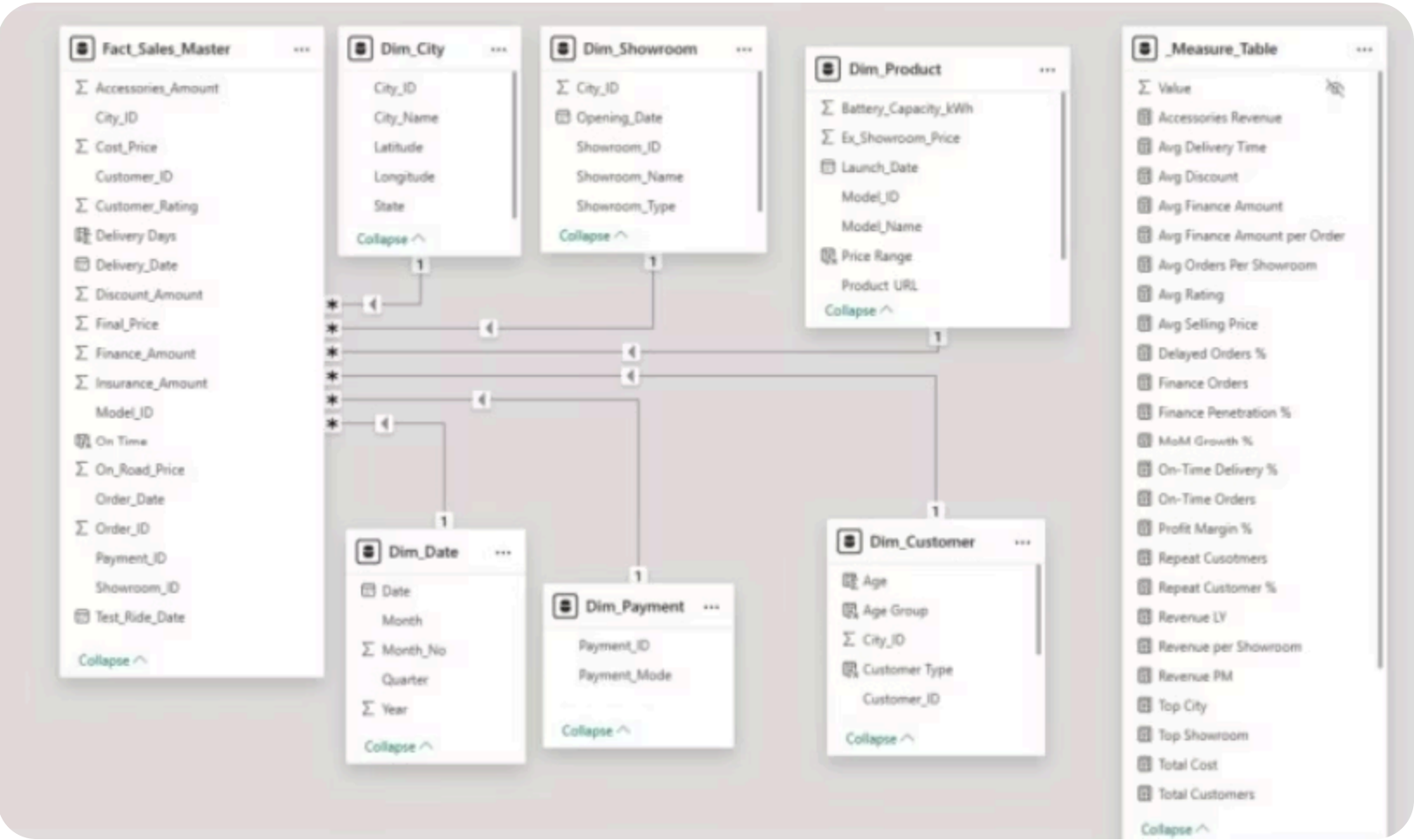
Data Model

Connected 1 Sales Table to 6 Detail Tables (Customer, Product, City, Store, Payment, Date)



Power BI

Created interactive charts and DAX formulas for dynamic reporting



Executive Dashboard — Business Overview

The dashboard consolidates all key performance indicators into a single, interactive view — enabling leadership to make fast, data-backed decisions across revenue, delivery, and operations.

₹11B

Total Revenue

₹2.28B

Total Profit

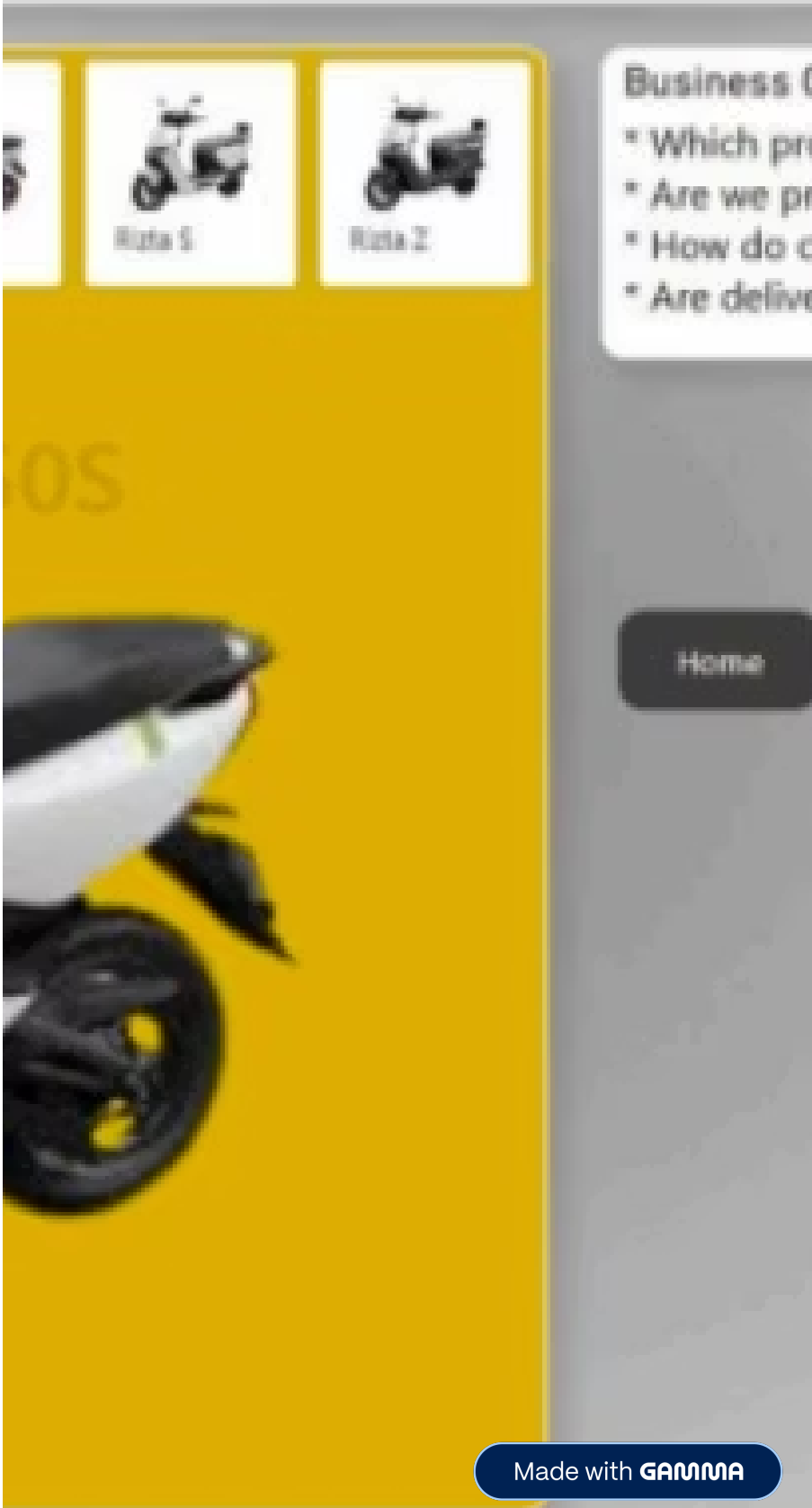
20.02% profit margin

71K

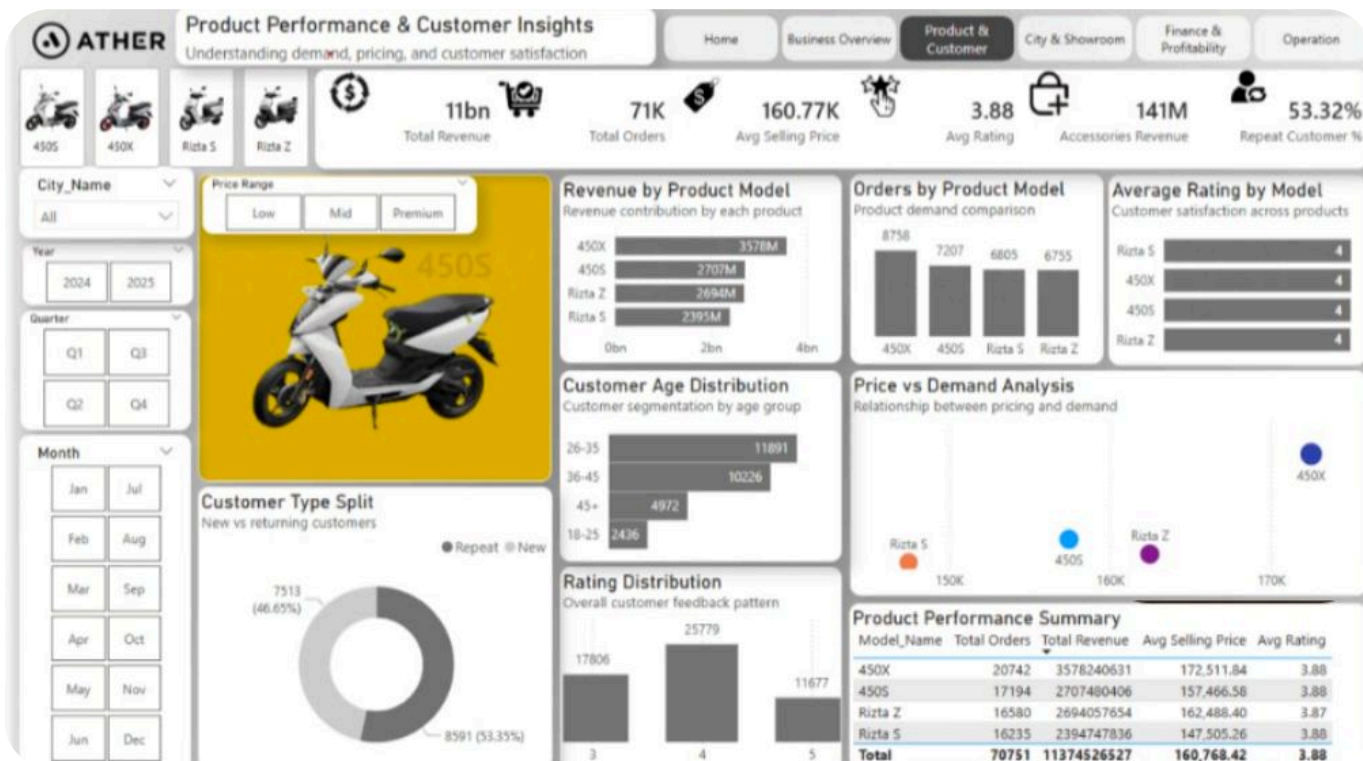
Vehicles Delivered

₹161K

Avg. Vehicle Price



Vehicle Models & Buyer Demographics



Top Seller: Ather 450X

Highest revenue model — **31.46%** of total sales (₹3,578M)

Primary Buyers

Young professionals aged **26–35 years** — 11,891 orders

Accessory Revenue

Generated **₹141M** extra through chargers and accessories

Customer Retention

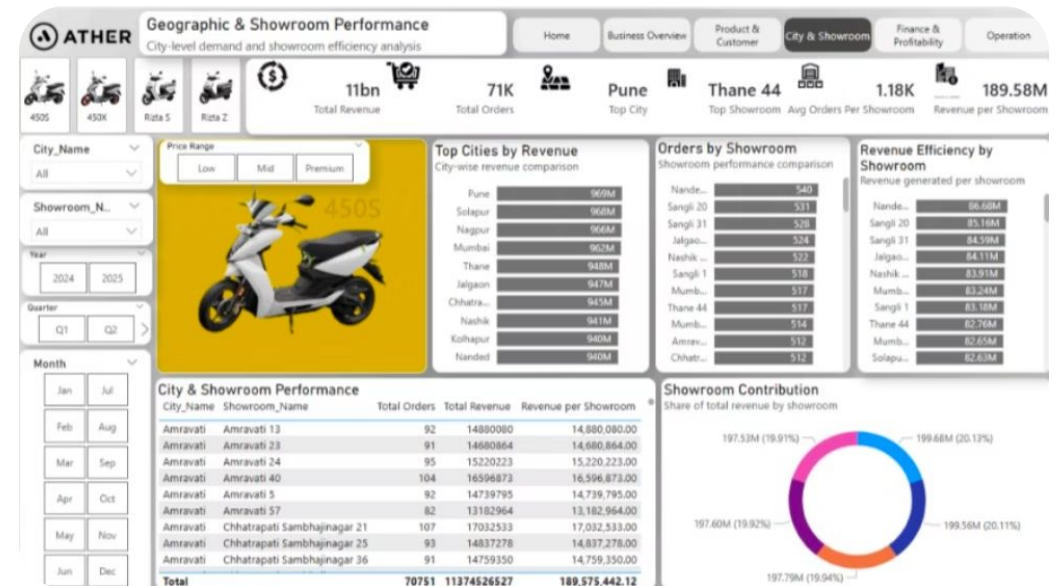
53.32% repeat customer rate — strong brand loyalty

City & Store Performance

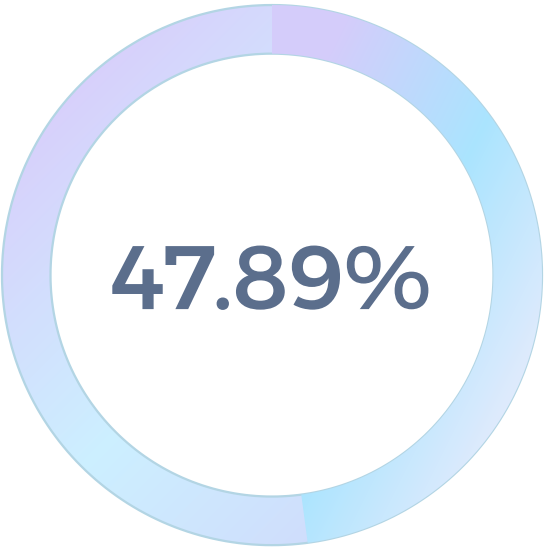
Regional Highlights

- **Top City:** Pune leads overall vehicle sales volume
- **Top Showroom:** Thane Experience Center 44 – highest revenue among all stores
- **Avg. Store Sales:** ₹189.58 Million per store
- **Market Spread:** ~20% per zone – strong expansion in smaller cities

❏ Even distribution across zones signals a healthy, scalable retail network beyond metro hubs.

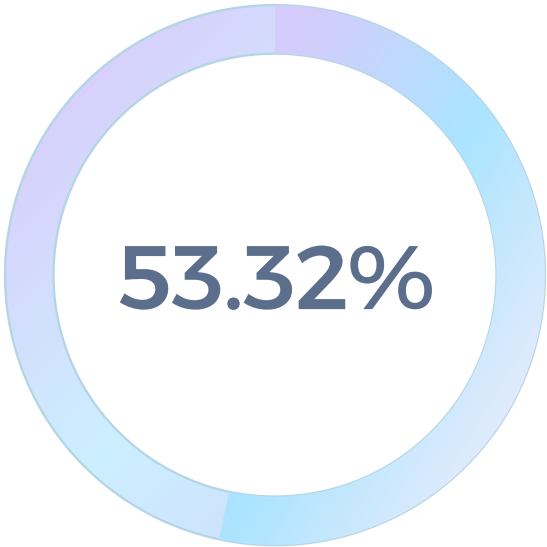


Customer Payments & Profit Margins



Loan Usage

Customers buying via Bank Loans or Credit Card EMIs



Repeat Customers

Returning buyers driving sustained revenue



Net Profit Margin

Solid margin even after discounts and operating costs



Average loan per vehicle stands at ₹81.46K, with total operating expenses controlled at ₹9.10 Billion – demonstrating disciplined cost management alongside strong financing adoption.

Delivery Timelines & Customer Satisfaction



Average Delivery Time

6.49 days from booking to vehicle delivery

On-Time Delivery Rate

62.79% of vehicles delivered within the promised window

Delayed Deliveries

37.21% face logistics delays – the primary area for improvement

Key Action

Fixing delays can improve customer rating from 3.88 → 4.5+

How I Can Drive Value For Your Organisation



Interactive Dashboards

Building simple, easy-to-use Power BI, Tableau, or Looker dashboards tailored to your business KPIs.



SQL & Data Pipelines

Setting up organised databases, clean data flows, and complex SQL queries for reliable reporting.



Advanced Analytics & Python

Automated data cleaning, Excel modelling, and statistical reporting to uncover hidden business insights.



Looking for a dedicated Data Analyst to solve reporting challenges or build custom business dashboards? Let's connect!
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